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Hyundai WIA Corp
Mr. Gyu Tae Bae
#107, Seongsanpaechongro, Seongsan-gu,
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Offer

Offers- No: 200905 / 3
date: 19/08/2022

Your technology contact

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Your Reference

Inquiry dated 20/07/2022
Contact Gyu Tae Bae
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EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 TWIN KBG for straight ball grooves (AC-, DO-joints) for the simultaneous hard turning and milling of 2 Inner Races of the same type

Inner Race HBJ#25
Inner Race HBJ#27
Inner Race HDJ#25
Inner Race HDJ#27



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Price does not include options (*)

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Managing Directors:
Dr. Guido Hegener, Dr. Mathias Klein
Michael Lanik, Sven Hartwich
Commercial Register Ulm: HRB 533796
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Bayerische Landesbank München 'IBAN DE59 7005 0000 0004 2569 67' BIC BYLADEMMXXX



Price Summary

1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 TWIN KBG

1.1 BASIC MACHINE

1.1.16 1 PRICE OF MACHINE VSC315TWIN KBG (SIEMENS) 609.600,00 €

1.2 ADDITIONAL EQUIPMENT

1.2.2	2	TOOL BLOCK FOR 1 TURNING TOOL CAPTO C6 manually clamped	30.800,00 €
1.2.5	1	LIGHT BEAMS FOR LOADING CONVEYOR	950,00 €
1.2.6	1	LINEAR GANTRY LP3, WITH ROTATION MODULE	23.300,00 €
1.2.7	1	GRIPPER JAWS INNER RACE CLAMPING DIA. 19-32	0,00 €
1.2.8	1	POSITION DETECTION	2.550,00 €
1.2.9	1	SHUTTLE ZR120TWIN R-L FOR WORKPIECE LOADING	34.800,00 €
1.2.10	2	WORKPIECE CARRIER RIGID INNER RACE - HBJ#25	3.700,00 €
1.2.11	2	WORKPIECE CARRIER RIGID INNER RACE - HBJ#27	3.700,00 €
1.2.12	2	WORKPIECE CARRIER EXPANDABLE INNER RACE - HDJ#25	7.900,00 €
1.2.13	2	WORKPIECE CARRIER EXPANDABLE INNER RACE - HDJ#27	7.900,00 €
1.2.14	2	AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK	2.400,00 €
1.2.15	1	SHUTTLE ZR120TWIN R-L FOR WORKPIECE UNLOADING	31.800,00 €
1.2.16	1	PRISM FINISH PART FOR INNER RACES	625,00 €
1.2.17	1	LIGHT BEAM FOR UNLOADING CONVEYOR	900,00 €
1.2.19	4	SAFETY SWITCH WITH CABLE FOR SAFETY ENCLOSURE	5.200,00 €
1.2.21	1	2 SWARF CONVEYORS, CHAIN TYPE	27.400,00 €
1.2.22	1	CHIP BOX (LARGE VERSION) - SUPPLIED BY CUSTOMER	0,00 €
1.2.23	1	AIR GUN	1.050,00 €
1.2.24	1	DRY-DUST AND MIST EXTRACTOR	5.250,00 €
1.2.26	1	EMAG - REMOTE DIAGNOSIS	1.650,00 €
1.2.27	1	NC-GUIDED EMERGENCY RETRACTION AT POWER FAILURE (ESR)	10.500,00 €
1.2.28	1	NETWORK DRIVE CONNECTION	2.000,00 €
1.2.29	2	INTERFACE TO EXTERNAL HANDLING SYSTEM - HARDWARE	7.400,00 €

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1.2.30	1	MINI MANUAL CONTROL UNIT (SIEMENS)	4.500,00 €
1.2.31	1	CONNECTING KIT FOR MINI HAND-HELD CONTROL UNIT – SIEMENS	825,00 €
1.2.32	1	WORKSHOP-ORIENTATED PROGRAMMING DO-JOINTS	6.000,00 €
1.2.33	1	WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS	6.000,00 €
1.2.35	1	ADDITIONAL ADAPTION (TROPICAL PACKAGE),	12.600,00 €
1.2.37	1	3-COLOUR STATUS LIGHT (Q-LIGHT)	1.200,00 €
1.2.38	1	PNEUMATIC PRESSURE BOOSTER	1.000,00 €
1.2.39	1	AIR DRYER FOR COMPRESSED AIR SUPPLY	5.800,00 €
1.2.40	1	SPECIAL MACHINE COLOUR	3.450,00 €

1.3 COMPONENT RELATED EQUIPMENT

1.3.2	1	SET OF TURNING TOOLS FOR HARD TURNING	9.250,00 €
1.3.3	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR HBJ#25	13.900,00 €
1.3.4	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR HBJ#27	13.900,00 €
1.3.5	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR HDJ#25	13.900,00 €
1.3.6	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR HDJ#27	13.900,00 €
1.3.7	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE	1.950,00 €
1.3.8	1	TEST BARS HSK63-C AND CAPTO C6	2.500,00 €
1.3.9	1	SETTING DIE FOR MILLING HEAD EXCHANGE	825,00 €
1.3.10	1	FIXING BOLT / GUIDE FOR MILLING /TURNING TOOL HOLDER - SPARE	250,00 €
1.3.12	2	ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIN85 - with 1 medium duct	10.500,00 €
1.3.13	2	AIR SENSING	10.800,00 €
1.3.15	2	ADAPTER FOR SPINDLE (CENTROTEX)	22.400,00 €
1.3.17	2	MANDREL DIA. 22,0 mm - 29,0 mm (CENTROTEX)	26.200,00 €
1.3.19	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR HBJ#25	9.200,00 €
1.3.20	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR HBJ#27	9.200,00 €
1.3.21	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR HDJ#25	9.200,00 €
1.3.22	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR HDJ#27	9.200,00 €

1.4 SERVICES

1.4.2	1	PROJECT ENGINEERING	37.100,00 €
1.4.3	1	TOOLING LAYOUT FOR INNER RACE HBJ#25	6.300,00 €

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1.4.4	1	TOOLING LAYOUT FOR INNER RACE HBJ#27	6.300,00 €
1.4.5	1	TOOLING LAYOUT FOR INNER RACE HDJ#25	6.300,00 €
1.4.6	1	TOOLING LAYOUT FOR INNER RACE HDJ#27	6.300,00 €
1.4.7	1	NC-PROGRAM TO MACHINE FOR INNER RACE BJ23	1.500,00 €
1.4.8	1	NC-PROGRAM TO MACHINE FOR INNER RACE BJ24	1.500,00 €
1.4.9	1	MACHINE SETUP FOR INNER RACE HBJ#25	7.900,00 €
1.4.10	1	MACHINE SETUP FOR INNER RACE HBJ#27	7.900,00 €
1.4.11	1	MACHINE SETUP FOR INNER RACE HDJ#25	7.900,00 €
1.4.12	1	MACHINE SETUP FOR INNER RACE HDJ#27	7.900,00 €
1.4.13	1	PRE-ACCEPTANCE OF THE MACHINE WITH INNER RACE HBJ#25	9.200,00 €
1.4.14	1	PRE-ACCEPTANCE OF THE MACHINE WITH INNER RACE HDJ#25	9.200,00 €
1.4.15	1	FUNCTION TEST OF THE MACHINE WITH INNER RACE HBJ#27	3.350,00 €
1.4.16	1	FUNCTION TEST OF THE MACHINE WITH INNER RACE HDJ#27	3.350,00 €
1.4.17	1	INSTALLATION AND DEMONSTRATION (OVERSEAS)	35.000,00 €
1.4.18	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE	5.500,00 €
1.4.19	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE	5.500,00 €
1.4.20	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE	5.500,00 €
1.4.21	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE	5.500,00 €
1.4.22	1	PRODUCTION SUPPORT 1 WEEK AFTER START OF PRODUCTION	9.100,00 €
1.4.27	4	ADDITIONAL SET OF DOCUMENTATION	4.000,00 €
1.4.28	2	DOCUMENTATION ON USB	250,00 €
1.4.32	1	COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS	11.300,00 €
1.4.34	1	OPERATING COURSE (Basic training)	6.125,00 €
1.4.35	1	MAINTENANCE (MECHANICAL AND ELECTRICAL) SERVICE COURSE (Basic training)	6.125,00 €
1.4.37	1	PACKING AND TRANSPORTATION OF TEST PIECES - AIR FREIGHT	5.800,00 €
1.4.38	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - KOR --> GER	10.200,00 €
1.4.39	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - GER --> KOR	10.200,00 €
1.4.40	1	FOB HAMBURG	19.600,00 €
Total price:			1.301.525,00 €
Unique special discount:			-371.525,00€

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1.5 MACHINE SPECIFICATIONS

Final price:

930.000,00 €

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1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 TWIN KBG

1.1 BASIC MACHINE

- 1.1.1 1 MACHINE BED
The swivelling axis with housing for the milling spindle is integrated in the machine bed. The machine bed manufactured from high quality Mineralit (cast granite) guarantees superior thermal stability and excellent dampening characteristics. Mineralit has excellent vibration dampening properties which are six to eight times better than cast iron. The large distance between the two high precision pre-stressed linear roller guideways of the X-axis guarantees high accuracy and high dynamics. The guideways are outside the working area and therefore require no protective covers.
- 1.1.2 1 MACHINING AREA
The machining area is effectively separated from the load/unload area by a partition wall facing the pick-up station.
A large door at the front of the machine provides access to the machining area. The door features a safety glass window to DIN EN ISO 23125:2010 specification.
A inspection lamp is installed in the machining area.
- 1.1.3 1 MULTI-FUNCTIONAL CROSS-SLIDE
This unit is equipped with two separate axes (X and Z) per spindle. Both spindles are controlled independently by the CNC to achieve separate tool correction. High rapid feed enable short idle times by component load/unload and gauging.
The axes drives are quick reaction frequency controlled service free AC motors which drive the slides over high precision ballscrew spindles. The X- and Z-axis are equipped with completely enclosed linear measuring systems.
On activation of the emergency-off switch or in case of power failure a feed motor with integrated brake will stop the vertical axis.
- 1.1.4 2 MOTOR SPINDLES
Each spindle headstock with motor spindle.
The motor spindles are a high dynamic frequency controlled service free AC asynchronous spindle motors.
High degree of spindle rigidity due to triple bearing support over a short distance.
Precision shoulder bearings arranged in tandem (double-O). Additional bearing at spindle end for length compensation. All bearings are lifetime lubricated and thus maintenance free.
Sealing air to prevent dirt from penetrating from the outside.
The guideway system in the Z axis is equipped with hydrostatic bearings for compensation the cutting forces which guarantee excellent guideway quality by minimum friction.
Thermo-symmetrical design and integrated coolant system guarantee constantly maintained accuracy.
Positioning accuracy of spindle: +/- 0,01 angular degrees (C-axis)
Electro-hydraulic spindle locking optionally.

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- 1.1.5 1 **SWIVEL UNIT WITH MOTOR SPINDLE 8500 RPM FOR STRAIGHT BALL GROOVES**
Housing running in cylindrical roller bearings supported in the sidewalls of the machine base.
Swivel angle -30° to +70°
Driven by torque drive. Direct position feedback system.
- 2 MOTOR SPINDLES
- | | | |
|---------------------------------|-------|-------|
| Max. spindle speed | rpm | 8.500 |
| Bearing dia. | mm | 80 |
| Power rating at 100% duty cycle | kW | 15 |
| Full power at speed of | rpm | 3.000 |
| Tool register | HSK-C | 63 |
- Fluid-cooled spindle housing. Precision hybrid ceramic bearings. In-built thermistor motor protection. Rotating components precision balanced on two planes. Tool holder system with locating face on shoulder. Torsion-proof. Tools mechanically clamped. (Mapal system)
(Motor spindles: warranty period 12 months)
- 1.1.6 1 **ENERGY CONTAINER WITH CONTROL CABINET**
Equipped to VDE 0113 and EN 60204, part 1 and integrated in the over-all machine frame, thus ensuring a totally transportable unit. 400 V machine voltage, 230 V control voltage (AC)/24 V DC.
The power supplies of the cross slides and main spindles are totally outside the machining and swarf area. Easy access to the axis drives, recirculating ballscrews etc. is possible through covers in the side walls or from the top of the machine.
- 1.1.7 1 **HYDRAULIC UNIT**
The hydraulic unit is equipped with a self regulating pump.
- 1.1.8 1 **CENTRAL GREASE LUBRICATION SUPPLY SYSTEM**
for lubrication of the recirculating ballscrews and guideway systems.
- 1.1.9 1 **SERVICE UNIT FOR COMPRESSED AIR SUPPLY**
- 1.1.10 1 **COOLING UNIT (water - air)**
A cooling unit controls the thermal stability of the machine. Spindle bearings, drive unit, motor for milling spindle and machine base are maintained at ambient temperature (10°C - 40°C). The electrical cabinet is cooled by an air conditioner in the control cabinet door.
- 1.1.11 1 **CNC-CONTROL SIEMENS**
According to control specification.
- 1.1.12 1 **EMAG OPERATING AND SETTING MASKS**
The EMAG masks offer the setter/ operator optimal support in setting and operating the machine.
Multiple base- and monitoring functions including data input are, depending on the machine type, available.
- 1.1.13 1 For simple and fast programming of the machine on an appropriate ball track, the machine can be equipped with workshop-oriented programming modules. These modules are not included in the basic machine price and are described separately in the chapter „options to the CNC control“.
On the other hand, the part programs can also be completed by the operator in normal ISO code.

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1.1.14	1	STANDARD COLOUR Machine - light grey to RAL 7035, front door - grey to RAL 7046, automation, swarf conveyor - grey to RAL 7046.	
1.1.15	1	NOISE LEVEL A maximum average corrected sound level of 78 dB(A) to DIN 45635 is obtained when turning with a continuous cut on a C45 steel disc in both longitudinal and transverse direction. If the workpiece has for example an interrupted cut due to its geometry or its clamping condition there may occur a higher sound level. For an extra cost sound deadening material can be fitted to reduce this higher sound level.	
1.1.16	1	PRICE OF MACHINE VSC315TWIN KBG (SIEMENS)	609.600,00 €

1.2 ADDITIONAL EQUIPMENT

1.2.1 TOOLING SYSTEM

1.2.2	2	TOOL BLOCK FOR 1 TURNING TOOL CAPTO C6 manually clamped consisting of: toolholder with clamping insert C6 With internal air blast feeding to the tools. The coolant system guarantees constantly maintained accuracy.	30.800,00 €
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1.2.3 COMPONENT STORAGE AND TRANSPORT SYSTEM

1.2.4	1	WORKPIECE LOADING by motorised chain-type conveyor (supplied by customer)	
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Please note:

The height of the infeed conveyor have to be 1050 [mm].

The infeed conveyor should be provided with height-adjustable feet.

The workpiece stop at the end of the infeed conveyor must be adjustable too.

1.2.5	1	LIGHT BEAMS FOR LOADING CONVEYOR 1x storage position free / 1x loading conveyor full For external loading	950,00 €
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1.2.6	1	LINEAR GANTRY LP3, WITH ROTATION MODULE 2-axis linear gantry Drive by NC servomotor. Direct connection to the machine. Energy guiding on all axes via cable track. Rotation unit driven by a servo motor with rotary inlet. Workpiece transport with a pneumatic 3-finger centric gripper incl. 1 set of workpiece-specific gripper jaws. without position detection and sensor gripper monitoring. Prepared for fitting of an automatic alignment device for the correct angular positioning of workpieces in the ball tracks (hard machining).	23.300,00 €
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Swivel angle	± 360°
X-axis stroke	630 mm
Z-axis stroke	500 mm
Workpiece weight	max. 3,5kg

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1.2.7	1	GRIPPER JAWS INNER RACE CLAMPING DIA. 19-32 1 set of gripper jaws for 3-finger centric part gripper	
		Included in pos. "LINEAR GANTRY LP3, WITH ROTATION MODULE".	
1.2.8	1	POSITION DETECTION Position detection of the workpiece via Keyence laser sensor, mounted on the gantry frame LP3	2.550,00 €
1.2.9	1	SHUTTLE ZR120TWIN R-L FOR WORKPIECE LOADING for the workpiece transport through the machine. With two basic pallets for the un machined components. Consisting of two spring loaded pickup stations, including detection and emergency stop Slide-/brake plate topped with spring steel, a pan for dry machining. Mounting kit on the machine base, 1x light beam make Balluff. Driven by servomotor The pick-up spindle picks up the un machined components from the shuttle and deposits the machined components onto the unloading conveyor. The components are automatically positioned for pick-up by the pick-up spindle. Various component diameters and heights are defined in the NC program.	34.800,00 €
1.2.10	2	WORKPIECE CARRIER RIGID INNER RACE - HBJ#25 Hard machining for the location of inner races. Designed for one workpiece.	3.700,00 €
1.2.11	2	WORKPIECE CARRIER RIGID INNER RACE - HBJ#27 Hard machining for the location of inner races. Designed for one workpiece.	3.700,00 €
1.2.12	2	WORKPIECE CARRIER EXPANDABLE INNER RACE - HDJ#25 for the radial location of inner races in the angular ball tracks by spring-actuated balls. Designed for one workpiece.	7.900,00 €
1.2.13	2	WORKPIECE CARRIER EXPANDABLE INNER RACE - HDJ#27 for the radial location of inner races in the angular ball tracks by spring-actuated balls. Designed for one workpiece.	7.900,00 €
1.2.14	2	AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK For the cleaning of the pick-up unit and the clamping chuck by air blast. The air is processed by a service unit and directed at the component through nozzles.	2.400,00 €
1.2.15	1	SHUTTLE ZR120TWIN R-L FOR WORKPIECE UNLOADING for the workpiece transport through the machine. With two basic pallets for the un machined components. Consisting of two spring loaded pickup stations, including detection and emergency stop Slide-/brake plate topped with spring steel, a pan for dry machining. Mounting kit on the machine base, 1x light beam make Balluff. Driven by servomotor The pick-up spindle picks up the un machined components from the shuttle and deposits the machined components onto the unloading conveyor. The components are automatically positioned for pick-up by the pick-up spindle. Various component diameters and heights are defined in the NC program.	31.800,00 €
1.2.16	1	PRISM FINISH PART FOR INNER RACES Prism for pushing the finished parts onto the outstocker.	625,00 €
1.2.17	1	LIGHT BEAM FOR UNLOADING CONVEYOR 1 x workpiece at the unloading position For external unloading	900,00 €

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Hyundai WIA Corp

19/08/2022

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1.2.18 1 SAFETY ENCLOSURE (supplied by customer)

Please note:

- **EMAG will supply an incomplete / unsafe machine.**
- **The customer is responsible for the safety of the system (machine and periphery).**

1.2.19 4 SAFETY SWITCH WITH CABLE FOR SAFETY ENCLOSURE 5.200,00 €
for customer supplied safety enclosure
consisting of:
- cable
- closing mechanism

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1.2.20 COOLANT AND SWarf DISPOSAL SYSTEM

1.2.21	1	2 SWarf CONVEYORS, CHAIN TYPE Consisting of two single swarf conveyors: The first swarf conveyor is mounted laterally on the unloading side of the machine. Swarf exit height 350 mm The second swarf conveyor carries the swarfs to the rear side of the machine. Swarf exit height 1250 mm.	27.400,00 €
1.2.22	1	CHIP BOX (LARGE VERSION) - SUPPLIED BY CUSTOMER with caster	
1.2.23	1	AIR GUN with hose	1.050,00 €
1.2.24	1	DRY-DUST AND MIST EXTRACTOR For dry hard machining. Including stainless steel separator baffle. High-precision filtration unit with filter inserts. Air volume 1000 m3/h.	5.250,00 €

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**1.2.25 OPTIONAL EQUIPMENT FOR CNC CONTROL**

1.2.26	1	EMAG - REMOTE DIAGNOSIS Control system: Sinumerik 840Dsl / Fanuc 3xi - Linkage with user network necessary (DHCP preferred, don't use IP-Segment 192.168.1.0/24 and 192.168.214.0/24) - Connection via Internet and safe VPN technology to EMAG server Scope of supply: - Software for remote maintenance - Router in the electrical cabinet (incl. firewall) Content: - Remote access to NC / PLC / operating system for detailed error diagnosis and the possibility to support the customer in case of queries or problems. Preconditions: - Internet connection of the machine (only during the remote diagnosis meeting) to a public IP address of Fa. EMAG Your advantages: - Quick, cost-effective support, resulting in reduced downtimes. - Up-to-date problem solving without lengthy delays. - Program extensions can be realized at a reduced cost. - "On-the-spot" co-operation between specialist and operating/service personnel allows for complex errors to be diagnosed, analysed and eliminated quickly. Availability: The service is available at the local service times for the following countries: Europe (excluding Russia), China and USA. For all other countries the service is linked to the service times of the CET. Free of charge during the warranty period, thereafter at the current hourly service rates.	1.650,00 €
1.2.27	1	NC-GUIDED EMERGENCY RETRACTION AT POWER FAILURE (ESR) Interpolated retraction of axis in case of power failure, voltage fluctuation and drive failure. Electric components (condenser modul) are used which maintain the voltage for the duration of the retreat and use the energy for the nc-guided retract of the axes. (strongly recommended for unstable power supply, expensive tools or workpieces)	10.500,00 €
1.2.28	1	NETWORK DRIVE CONNECTION (Control system Siemens 840Dsl) Linking the machine control to your network after startup in your factory, so that network drives are available in the user interface. NC-programs or any data can be stored or loaded from there. Customer-specific requirements / restrictions for network integration: - Operation with DHCP server yes / no; if no, then IP address, sub-net mask and default gateway - DNS server and enable code of network source (also IP-address of server, if DNS not known) - User name and password for the automatic preparation of the connection - Network cable into the electrical cabinet When providing server and directory names it has to be taken into account that the 840Dsl features an English language Windows operating system that cannot handle the German umlauts ä, ö and ü. When installing of security update, the company Siemens restrictions are to consider. If in the domain automatically software is installed on all computers, so the functionality is no longer guaranteed after restart!	2.000,00 €

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1.2.29	2	INTERFACE TO EXTERNAL HANDLING SYSTEM - HARDWARE Safety signal interface via HAN24E plug on electrical cabinet with 8 potential-free inputs and 8 potential-free outputs for the change of safety signals "emergency-stop" and "access door open" Handling-signal interface via HAN16E plug on electrical cabinet Deviations of the standard interface on request at additional costs. (1.000,00 € / signal)	7.400,00 €
1.2.30	1	MINI MANUAL CONTROL UNIT (SIEMENS) For operating a combined positioning and transfer unit and/or a linear gantry loader. In the setting up mode of the automation components: the shuttle, the vertical and horizontal axis of the gantry, also the swivelling and gripping function of the part gripper can be manually operated with the manual control unit. If the doors of the safety enclosure are open, the acknowledgment key must be pressed on the manual control unit.	4.500,00 €
1.2.31	1	CONNECTING KIT FOR MINI HAND-HELD CONTROL UNIT – SIEMENS	825,00 €
1.2.32	1	WORKSHOP-ORIENTATED PROGRAMMING DO-JOINTS For inner races (straight grooves). For quick and easy programming parameters for componentrelated dimensions are adapted. Same process for tool-offset. Every tool is allocated an empirically determined preset quantity of workpieces to be machined. On expiry of the tool life the machine is stopped. Control monitor display: tool-no., preset and remaining tool life quantity of workpieces per tool.	6.000,00 €
1.2.33	1	WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS For inner races (straight grooves). For quick and easy programming parameters for componentrelated dimensions are adapted. Same process for tool-offset. Every tool is allocated an empirically determined preset quantity of workpieces to be machined. On expiry of the tool life the machine is stopped. Control monitor display: tool-no., preset and remaining tool life quantity of workpieces per tool.	6.000,00 €

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**1.2.34 COUNTRY SPECIFIC EQUIPMENT**

1.2.35	1	ADDITIONAL ADAPTION (TROPICAL PACKAGE) for max. 48°C ambient temperature on machines with "water /air" cooling unit for max. 52°C ambient temperature on machines with "water /water" cooling unit for min. 10°C with grease lubrication at max. 90% humidity air condition in door of electrical cabinet additional 230V transformer for main switch electrical cabinet heated when machine switched off operating console heated when machine switched off additional fan, for air circulation in the operator console	12.600,00 €
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1.2.36 OTHER AUXILIARY EQUIPMENT

1.2.37	1	3-COLOUR STATUS LIGHT (Q-LIGHT) mounted on machine. Colour code: red: machine stopped yellow: warning -tool to be changed, -lubricant running low, etc. green: machine in automatic cycle.	1.200,00 €
1.2.38	1	PNEUMATIC PRESSURE BOOSTER to increase air pressure from 4 bar to 6 bar. (only for part grippers) Maker: Festo Type: DPA-63-10	1.000,00 €
1.2.39	1	AIR DRYER FOR COMPRESSED AIR SUPPLY Absorption air dryer for dry and clean air. Manufacturer: Festo Type: PDAD-51-G1/2 incl. combined maintenance unit	5.800,00 €
1.2.40	1	SPECIAL MACHINE COLOUR Complete machine according to customers colour specification.	3.450,00 €

1.3 COMPONENT RELATED EQUIPMENT**1.3.1 TOOL EQUIPMENT**

1.3.2	1	SET OF TURNING TOOLS FOR HARD TURNING consisting of: - 4 turning tool holder Capto C6 - 40 cbn cutting inserts	9.250,00 €
1.3.3	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR HBJ#25 Head system with 4 brazed cbn cutting inserts consisting of: - 4 basic tool holder HSK63-C - 30 cutting heads Ø15,2	13.900,00 €

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1.3.4	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR HBJ#27 Head system with 4 brazed cbn cutting inserts consisting of: - 4 basic tool holder HSK63-C - 30 cutting heads Ø16,8	13.900,00 €
1.3.5	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR HDJ#25 Head system with 4 brazed cbn cutting inserts consisting of: - 4 basic tool holder HSK63-C - 30 cutting heads Ø14,287	13.900,00 €
1.3.6	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR HDJ#27 Head system with 4 brazed cbn cutting inserts consisting of: - 4 basic tool holder HSK63-C - 30 cutting heads Ø15,081	13.900,00 €
1.3.7	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE consisting of: - 1 cone wiper - 1 torque wrench with insert - 2 plugs HSK63 - 1 ball plug gauge HSK63	1.950,00 €
1.3.8	1	TEST BARS HSK63-C AND CAPTO C6 for checking the machine geometry Consisting of: 1x aluminum case 1x foam insert 1x test bar HSK63-C 1x test bar Capto C6	2.500,00 €
1.3.9	1	SETTING DIE FOR MILLING HEAD EXCHANGE consisting of : 1x fixture for milling head change for mounting on a bench.	825,00 €
1.3.10	1	FIXING BOLT / GUIDE FOR MILLING /TURNING TOOL HOLDER - SPARE consisting of: - 2x fixing bolt for turning tool holder as spare - 2x fixing guide for turning tool holder as spare - 2x fixing bolt for milling tool holder as spare	250,00 €

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**1.3.11 CHUCKING SYSTEM**

- 1.3.12 2 ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIN85 10.500,00 €
- with 1 medium duct
Max. drawbar force 3.325 daN at 70 bar.
The clamping pressure is manually infinitely adjustable. .
The clamping cylinder stroke and the clamping pressure are controlled by a linear measuring system and by a pressure switch.

- 1.3.13 2 AIR SENSING 10.800,00 €
The air sensing is exclusively used to identify the workpiece contact on the clamping device.
This is done by measuring the dynamic pressure that occurs between the workpiece and the clamping device.
The air is supplied by one or more nozzles (depending on clamping fixture type).
The pressure is measured with a pressure switch.
The air supply is controlled by a directional valve.
The air flow is sent through the main spindle.

NOTE:

Requires CHUCKING CYLINDER WITH MEDIUM DUCT

Equipping the clamping fixture with a pneumatic system monitor is not included in the price of this item.

1.3.14 HAINBUCH - ADAPTER FOR SPINDLE (QUICK CHANGE)

- 1.3.15 2 ADAPTER FOR SPINDLE (CENTROTEX) 22.400,00 €
for quick change of the clamping mandrels / collet chucks
Prepared for air sensing.
The adaption of the clamping device to the draw bar is via a bayonet.

1.3.16 HAINBUCH - MANDREL ROUND - CENTROTEX CONNECTION

- 1.3.17 2 MANDREL DIA. 22,0 mm - 29,0 mm (CENTROTEX) 26.200,00 €
For clamping of Inner Races.
Installation by CentroTex-interface for quick change of the mandrel (spindel adapter is required).
Round clamping geometry.
Clamping range: dia. 22,0 mm - 29,0 mm
Prepared for air sensing.
Without workpiece-specifically interchangeable parts.

1.3.18 HAINBUCH - MANDREL ROUND - INTERCHANGEABLE PARTS

- 1.3.19 2 SET OF INTERCHANGEABLE PARTS (MANDREL) - IR HBJ#25 9.200,00 €
Laid out for one inner race.
Consisting of collet, part stop and in total 1x control ring.
- 1.3.20 2 SET OF INTERCHANGEABLE PARTS (MANDREL) - IR HBJ#27 9.200,00 €
Laid out for one inner race.
Consisting of collet, part stop and in total 1x control ring.
- 1.3.21 2 SET OF INTERCHANGEABLE PARTS (MANDREL) - IR HDJ#25 9.200,00 €
Laid out for one inner race.
Consisting of collet, part stop and in total 1x control ring.

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1.3.22	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR HDJ#27 Laid out for one inner race. Consisting of collet, part stop and in total 1x control ring.	9.200,00 €
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1.4 SERVICES

1.4.1	1	Please note that no additional services are contained in the previously specified prices. We would be pleased to carry out the following services and would quote according to your wishes. - Preparation of tooling layouts and NC programs - Set-up machine for component trials on your specific workpiece in our works - Machine approval in our works by your engineers. - Final acceptance at customer's plant with a specific workpiece	
1.4.2	1	PROJECT ENGINEERING Project engineering and coordination with suppliers and customers. Engineering investigations (machine set up, with central automation, central exhaust, chip disposal) Customization of interfaces with the suppliers.	37.100,00 €
1.4.3	1	TOOLING LAYOUT FOR INNER RACE HBJ#25 Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments, for one workpiece.	6.300,00 €
1.4.4	1	TOOLING LAYOUT FOR INNER RACE HBJ#27 Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments, for one workpiece.	6.300,00 €
1.4.5	1	TOOLING LAYOUT FOR INNER RACE HDJ#25 Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments, for one workpiece.	6.300,00 €
1.4.6	1	TOOLING LAYOUT FOR INNER RACE HDJ#27 Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments, for one workpiece.	6.300,00 €
1.4.7	1	NC-PROGRAM TO MACHINE FOR INNER RACE BJ23 Creating and importing the NC programm into the machine.	1.500,00 €
Please note: Set-up will be done by customer after signing final acceptance documents.			
1.4.8	1	NC-PROGRAM TO MACHINE FOR INNER RACE BJ24 Creating and importing the NC programm into the machine.	1.500,00 €
Please note: Set-up will be done by customer after signing final acceptance documents.			
1.4.9	1	MACHINE SETUP FOR INNER RACE HBJ#25 for one workpiece. Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	7.900,00 €

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1.4.10	1	MACHINE SETUP FOR INNER RACE HBJ#27 for one workpiece. Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	7.900,00 €
1.4.11	1	MACHINE SETUP FOR INNER RACE HDJ#25 for one workpiece. Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	7.900,00 €
1.4.12	1	MACHINE SETUP FOR INNER RACE HDJ#27 for one workpiece. Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	7.900,00 €
1.4.13	1	PRE-ACCEPTANCE OF THE MACHINE WITH INNER RACE HBJ#25 Specific workpiece: Inner Race HBJ#25 with measuring equipment supplied by customer. One workpiece in one clamping operation. Max. 30 components. Submission of proof to Cm of machine's capability of max. 3 measuring characteristics. For Cm of machine's capability measurements of workpiece 1, 5, 10, 15, 20, 25, 30.	9.200,00 €
1.4.14	1	PRE-ACCEPTANCE OF THE MACHINE WITH INNER RACE HDJ#25 Specific workpiece: Inner Race HDJ#25 with measuring equipment supplied by customer. One workpiece in one clamping operation. Max. 30 components. Submission of proof to Cm of machine's capability of max. 3 measuring characteristics. For Cm of machine's capability measurements of workpiece 1, 5, 10, 15, 20, 25, 30.	9.200,00 €
1.4.15	1	FUNCTION TEST OF THE MACHINE WITH INNER RACE HBJ#27 Specific workpiece: Inner Race HBJ#27 for one workpiece including test run, without SPC analysis, 10 parts machined in 100% drawing tolerance. With measuring equipment supplied by customer.	3.350,00 €
1.4.16	1	FUNCTION TEST OF THE MACHINE WITH INNER RACE HDJ#27 Specific workpiece: Inner Race HDJ#27 for one workpiece including test run, without SPC analysis, 10 parts machined in 100% drawing tolerance. With measuring equipment supplied by customer.	3.350,00 €

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1.4.17	1	INSTALLATION AND DEMONSTRATION (OVERSEAS) in your works	35.000,00 €
		The following services are included: - Machine installation and demonstration - Machine approval and hand-over - Talk-through of the machine documentation and service instruction - Operator instruction - Travel costs	
		Total of 15 man days	
		Not included are: - Machine set-up - Machine operation for production - Additional service	
		Additional service engineer costs for time in excess of that quoted will be charged extra.	
1.4.18	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE Specific workpiece: Inner Race HBJ#25 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces/workpiece spindle. Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.19	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE Specific workpiece: Inner Race HBJ#27 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces/workpiece spindle. Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.20	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE Specific workpiece: Inner Race HDJ#25 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces/workpiece spindle. Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.21	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE Specific workpiece: Inner Race HDJ#27 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces/workpiece spindle. Evaluation and guaranteed criteria are the same as for pre-acceptance test.	5.500,00 €
1.4.22	1	PRODUCTION SUPPORT 1 WEEK AFTER START OF PRODUCTION 1 week (5 working days) production support at customers plant Performed by EMAG India or EMAG Korea.	9.100,00 €
1.4.23	1	INSTALLATION POINT The installation point has to be prepared in accordance with the documentation supplied. We take it for granted that all conditions are met before our installation crew arrives. Where waiting times are incurred due to the conditions not having been met, these will be charged extra. During installation and commissioning our engineers have to be provided, on request, with the support of skilled customer personnel and similar assistance. This also applies to the operation of cranes, handling equipment and other aids.	

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- 1.4.24 1 **MACHINE LIFTING EQUIPMENT**
EMAG will provide a lifting equipment to lift the machine by crane, free of charge.
However, should the customer not return the equipment to EMAG within 4 weeks, freight
and charges prepaid, a separate invoice of EUR 1.500,00 will be issued.

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Offer : 200905 / 3

Hyundai WIA Corp

19/08/2022

21/33

Managing Directors:
Dr. Guido Hegener, Dr. Mathias Klein
Michael Lanik, Sven Hartwich
Commercial Register Ulm: HRB 533796
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1.4.25 DOCUMENTATION

1.4.26 1 DOCUMENTATION (Siemens)

OPERATING INSTRUCTIONS

1 complete set of operating instructions on paper

EU countries in the official language of the country; other countries on national regulations

Manuals for the machine:

- Introduction
- Safety instructions
- Product description
- Transport and installation
- Start-up
- Production
- Preventive maintenance

Manuals from the manufacturer of the control unit:

Siemens SIN 840D sl

- User manual
- Diagnostics manual
- Programming - Basics
- Programming - Cycles
- Programming - Operations planning and scheduling

Machine diagrams:

- Machine layout
- Transport diagram
- Machine data sheet
- Drive data sheet
- Connection data sheet for fluidics
- Connection data sheet for electrics

List of spare parts and wearing parts

Operating instructions from third parties

Documentation for clamping devices and tools

TECHNICAL DATA

1 set of technical data on paper, German or English

- Electrical circuit diagram with list of el. devices
- Fluid system diagram with list of el. devices
- Drawings of subassemblies with relevance for spare parts
- List of components with relevance for spare parts and wearing parts in relation to the drawings
- Technical data, test reports, declarations of incorporation from third parties

Other languages available on request for an additional charge; third party documentation excepted

1.4.27	4	ADDITIONAL SET OF DOCUMENTATION	4.000,00 €
		1 additional set of the documentation described above	

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1.4.28	2	DOCUMENTATION ON USB 1 complete set of documentation on USB	250,00 €
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1.4.29 CE-MARKING

- 1.4.30 1 WITHOUT CE MARKING / WITHOUT DECLARATION OF CONFORMITY (non-EU country, partly completed machinery)
The machinery is always designed and constructed in accordance with the applicable harmonized standards and safety instructions.
By request of the customer, the machinery is constructed as partly completed machinery (e.g. in the workpiece infeed and outfeed area) and is intended exclusively for incorporation into or assembly with other machinery or equipment.
The owner-user is responsible for securing these areas. The machinery may not be put into service until the owner-user has declared that the final machinery into which this partly completed machinery has been incorporated is in conformity with the regional provisions.

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1.4.31

CUSTOMER SPECIFICATION

1.4.32

1

COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS

11.300,00 €

Design of mechanical-, electrical- and fluid equipment according to "ENG_RFQ_for inner Race_WIA India Volvo GPA_WGT-M3-0019-001"

Due to our design we are unable to comply with the following items of your specification sheet, meaning we must apply for an exceptional permission.

Item	Customer requirement	EMAG commitment
Basic Condition	- AC 415V (+/-15%) x3 phase x 50Hz - Surrounding temperature -5°C - 50°C	- AC 415V (+/-10%) x3 phase x 50 Hz like in your specifications item Electric 1. Main Circuit. - With tropical package surrounding temperature between 10°C and 48°C
Noise level	less than 75 dB (A)	up to 80dB (A)
General specification of Hyundai WIA	Keep the "General specification of Hyundai WIA"	This document is not part of the inquiry and therefore it is not considered
Workpiece quality	Different requirements on pages 4 and 5 of the RFQ. Due to this, we unfortunately have to decline the requirements.	Regarding quality commitments please see attached quality data sheets. Only the EMAG quality data sheets are valid.
Main Circuit	Solenoid valve: 115V AC 50Hz, Lamp Type, Communication Type	24V DC with LED
CNC & PLC	- Voltage 240V - Memory USB Card - Memory EPROM	400V AC / 24V DC - Memory USB-Stick - No longer available from Siemens.
Safety fence	Need	Supplied by customer
CNC & PLC	Network interface - LAN	USB
Wiring	Remain the 30% marginal space and do not wire it hose together	max. 10% possible
Automation height	FL+950mm	FL+1030mm / FL+1050mm - Same as delivered with last MC's
Items for the capability (pre- and final acceptance)	Ball Track Milling: Groove Dia, Center, Ball Diameter	Ball diameter is no item for evaluation (Cm)

Price does not include options (*)

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1.4.33 TRAINING

- | | | | |
|--------|---|--|------------|
| 1.4.34 | 1 | <p>OPERATING COURSE (Basic training)
For two participants in your works, performed by EMAG India .
Matched to the machine design (modular customized)
Training course duration: 3 days / shift (shift monday - friday).
Language: English
Documentation: 4x English</p> | 6.125,00 € |
| 1.4.35 | 1 | <p>MAINTENANCE (MECHANICAL AND ELECTRICAL) SERVICE COURSE (Basic training)
For two participants in your works, performed by EMAG India.
Matched to the machine design (modular customized)
Training course duration: 3 days / shift (shift monday - friday).
Language: English
Documentation: 4x English</p> | 6.125,00 € |

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Offer : 200905 / 3

Hyundai WIA Corp

19/08/2022

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**1.4.36 TRANSPORT COSTS**

1.4.37	1	PACKING AND TRANSPORTATION OF TEST PIECES - AIR FREIGHT Transport (air freight) from customers plant in India (Chennai) to EMAG Maschinenfabrik in Salach. - 2x 300 test pieces of each acceptance part - 2x 150 test pieces of each function test part Max. total weight = 500 [kg] The customer has to take care of a transport-safe packaging in each case.	5.800,00 €
1.4.38	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - KOR --> GER Transport from customers plant in Korea to EMAG Salach Germany. (Air freight, max. 750kg) The customer has to take care of a transport-safe packaging in each case. It must be ensured that EMAG can reuse the packaging material.	10.200,00 €
1.4.39	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - GER --> KOR Transport from EMAG Maschinenfabrik Salach to customers plant in South Korea. (Air freight, max. 750kg) The customer has to take care of a transport-safe packaging in each case. It must be ensured that EMAG can reuse the packaging material.	10.200,00 €
1.4.40	1	FOB HAMBURG from Salach, Incoterms 2020 incl. seaworthy packing	19.600,00 €
Total price:			1.301.525,00 €
1.4.41	1	UNIQUE SPECIAL DISCOUNT	-371.525,00 €

1.5 MACHINE SPECIFICATIONS

Price does not include options (*)

Offer : 200905 / 3

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1.5.1 1 TECHNICAL MACHINE DATA VSC315TWIN KBG
Subject to change without notice

WORKING AREA

Chuck diameter	mm	250
Swing diameter	mm	260
X-stroke	mm	850
Z-stroke	mm	320

WORKPIECE DIMENSIONS

Workpiece length, max.	mm	250*
Workpiece O/D, max.	mm	120
Ball section dia., max.	mm	26

(*Longer workpieces on request)

MAIN SPINDLE

Spindle flange acc. to DIN 55026	Qty	2
Spindle front bearing diameter	size	8
max. spindle speed	mm	140
Spindle distance	rpm	3000
	mm	500

MAIN DRIVE

AC-Asynchronous motor 60/100 % ED	Qty	2
Max. power	kW	58/45
Full power at spindle speed	kW	58
Torque at 60/100 % ED	rpm	900
Max. torque at	Nm	620/480
	Nm	620

FEED DRIVE

Rapid traverse feed, X	m/min	45
Rapid traverse feed, Z	m/min	30
Feed force, X/Z	kN	11
Ball screw diameter X/Z	mm	50/40

MILLING SPINDLE

AC-motor	kW	15
Max. spindle speed	rpm	8500

ELECTRICAL EQUIPMENT according to VDE 0113

Supply voltage	V	400
Control circuit voltage - DC	V	24
- AC	V	230
Frequency	Hz	50
Total connected load	kVA	70
Input fuse	A	160

DIMENSIONS AND WEIGHT

Length	mm	4250
Width	mm	2800
Height	mm	3000
Weight	kg approx.	13000

1.5.2 1 SIEMENS OPERATOR PANEL 15" MULTITOUCH MONITOR
consisting of OP015 Black + IPC 427E + Control panel (MCP483C)

Price does not include options (*)

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- 1.5.3 1 **CNC CONTROL SIEMENS**
SINUMERIK 840 Dsl with integrated PLC S7-300
Operator software Sinumerik Operate with HMI-Pro (Windows 10)
Screen language and on-screen messages in German / English / in official EU language,
more languages available as an optional
CNC program and data memory 10 Mbyte
USB-interface (front) for memory-stick (USB 2.0)
Machine diagnostics via internet with VPN technology to verify EMAG's standard format
Software backup is to external media (USB stick integrated into IPC) to verify EMAG's
standard format
dataGrabber (Comara) – Capable of recording process data for diagnostics, optional
process optimization
No active evaluation possible on the control, storage of the data takes place in a ring
buffer

EMAG ADD-ON FUNCTIONS:

Machine-specific documentation of functions and parameters is available directly on the
control unit. (PDF display)
Automatic return to the home position upon machine error. Error recovery is a simple one
button process.
First run high speed traverse is at a reduced speed from home to the first cut. Reduced
speed allows you to avoid a collision and save damaged equipment and reprogramming
costs.
Internal measuring probe with collision monitoring on the traversing movements (Option)
Automatic workpiece gaging when switching to a replacement tool or tool insert (Option)
Automatic filling and emptying of the entire machine.
"Optional Stop" (M01) allows you to make a programmed stop or interrupt production after
finishing a part at a desired point.
Single cycle: Allows one part to be transferred and machined for setup and tryout
purposes.

STRUCTURED WORKPIECE PROGRAMMING:

Division of individual programs into subprograms, e.g. the program for the machining
contains only the technology program and not the loading part.
The loading program has individual parameters. Division into machine and work piece
specific parameters.
Programmable chuck with length measuring system in the pull rod.
Different clamping situations (Unclamped, Clamped and Over stroke) can be changed by
parameter assignment. Displayed by diagnostics menu.
Recording and displaying of the most important production data: work piece counter, shift
counter, cycle time, work piece status, operating hours.
Production of definable batch sizes. The machine stops upon reaching the selected no.
Optimized programming for the turret keeps chips from collecting.
Rinsing of clamping devices after a programmable number of parts
Number and positions of the workpiece carriers can be defined for closed automation.
Lifecycle screen that displays tool wear and tool life.

Final price:

930.000,00 €

Price does not include options (*)

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Commercial Conditions

CONFIDENTIALITY

Any commercial and operational written or oral information or data relating to this quotation / order, such as drawings, specifications, recommended machining data and clamping methods, etc., shall not be divulged to any third party (referring, in particular, but not exclusively, to competitors, for the purpose of bid comparisons) and are to be used solely within the meaning of this quotation / order.

The recipient of this quotation / order expressly accepts these obligations.

MACHINING

The determined cycle times result from values, which are based on data and experiences of acknowledged cutting material manufacturers. Changing rough part dimensions, tolerances, material as well as the material hardness are only possible during building and testing of the machine. Changes will only be accepted in written form and in mutual agreement.

PROVISION OF EQUIPMENT AND MATERIALS BY CUSTOMER

All tooling, work holding equipment, parts, etc. are to be supplied with DDP supplying plant EMAG Incoterms 2020. We wish to point out that we accept no responsibility for the functional capability and working order of any equipment provided.

The functional capability of the provided equipment is the responsibility of the buyer.

In the event of problems, we assure you of our constructive collaboration. Any additional costs arising from modification, optimization, etc. will be charged to the customer. Furthermore, we do not guarantee any quoted machining times in such cases.

When orders cannot be carried out on schedule, caused by material supplied by the customer or if technical problems should occur, the machine will be invoiced on the originally agreed delivery date, when the relevant payment will become due.

The agreed upon date of delivery date may be postponed to the time to achieve full operability, which shall not result in any event of default of the manufacturer.

DELIVERY PERIOD

Approx. 12 months FOB German Seaport when order will be placed until calendar week 34/2022. The delivery period is not binding and can change between the submission of quotation and placement of order.

Please discuss the subject of delivery with us at time of order placement and we will do everything possible to meet your requirements. Changes regarding the component/s or the machines or the order content after order placement may result in changes in price and delivery.

Compliance of delivery date requires timely down and / or partial payment by the customer within the prescribed payment terms. If receipt of down and or partial payment is delayed, the delivery shall be postponed accordingly, which shall not result in any event of default of the manufacturer.

DELIVERY BOTTLENECKS

If we are unable to meet delivery dates (non-availability of the service) for reasons for which we are not responsible (particular "Force Majeure"), we reserve the right to postpone the delivery date to a point in time at which the service is possible for us again. This applies for a maximum period of twelve months beyond the agreed delivery time. If the unavailability of the service will not end foreseeably, we reserve the right to withdraw from the contract in whole or in part.

A case of non-availability of the service also applies in particular to non-delivery by our suppliers (e.g. disrupted supply chains worldwide), unless EMAG is responsible for this circumstance through intentional or grossly negligent behavior.

We assure you of our endeavor to serve every order in a timely manner.

CONFORMITY WITH CUSTOMERS' REGULATIONS

has not been taken into account. Where unavoidable deviations from such regulations prove necessary, specific authorization will be requested.

PRICES

FCA (Free Carrier, Incoterms 2020). Packing, insurance and transport not included.

The payments are to be made without deduction. With delayed payment the relevant laws apply.

Price does not include options (*)

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Hyundai WIA Corp

19/08/2022

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VALIDITY OF PRICES

The prices quoted are valid for 60 days from the date of quotation. Orders placed after this period would be subject to a price increase of 1 % per month to cover increases in labour and material costs.

This quotation will replace all previous versions.

Price does not include options (*)

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TERMS OF PAYMENT

100 % of invoice value against irrevocable Letter of Credit payable as follows:
30 % of L/C-value after opening of the L/C as advance payment against simple receipt.
70 % of L/C-value at sight against shipping documents.
All L/C-charges are for buyer's account. L/C must be opened latest 30 days after order confirmation

RETENTION OF TITLE

The machine will remain the property of the builder until complete payment is received from the buyer. Upon placement of order the extended retention of title according to clause 48 of our General Conditions for Sales and Deliveries shall be deemed as agreed.

LIMITATION OF LIABILITY

The liability of the supplier is excluded for indirect damage, defects and consequential damage, or losses, e.g., loss of revenues, downtimes, production stoppages, capital costs, or costs incurred due to operational interruptions. This does not apply to damage willfully caused by the supplier, or in cases of statutory liability, in particular according to the Product Liability Act.

PERIOD OF LIABILITY

Any rights and claims of the buyer caused on defects shall be subject to a limitation period of twelve months. The limitation period starts with the final acceptance, but no later than 3 months from delivery, and does not apply to wear and tear parts.

WORK PIECES FOR MACHINE ACCEPTANCE

Both, pre-acceptance and final acceptance tests are carried out with either an EMAG test component to VDI standard, or the work piece specified by the customer's order.

The customer will supply an agreed number of work pieces for machining process optimisation and pre-acceptance tests. These work pieces must be from a single batch and of a quality that corresponds with the relevant manufacturing drawing. For preparatory work and preliminary machine acceptance the customer will, at the agreed time, supply EMAG free of charge with a sufficient number of sample work pieces of a quality that corresponds with the relevant manufacturing drawing. Any delay in the supply of these work pieces will result in the machine delivery being delayed and the starting date of a possible delayed machine delivery penalty being deferred accordingly.

FINAL ACCEPTANCE AND FINAL PAYMENT

Final acceptance tests are carried out as soon as the machine has been installed and commissioned. For this purpose, the buyer has to provide a sufficient number of work pieces, personnel and capacity to measure and evaluate the acceptance components. Once the machine is taken into production and the agreed component quality and cycle time have been achieved, final acceptance is considered to have taken place. Where final acceptance of the machine is not possible owing to reasons beyond EMAG's control (missing acceptance components, etc.), final contractual payment falls due immediately.

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TECHNICAL AVAILABILITY

Our confirmation of the technical availability according to VDI 3423 is based on following conditions:
Exclusively qualified operating personnel, monthly availability protocols addressed to our service manager, demonstrable preventive maintenance is carried out according to our machine documentation, connection to our tele service system if requested by our service department.
In case of non-compliance of these conditions, our commitment expires.

TECHNICAL DEVELOPMENT

In the interest of technical progress, we reserve ourselves the right to make design changes which are due to continuous product development.

RESERVATION CLAUSE

For offers within the European Union (without Germany):

„Notification of Intra-Community delivery according to Art. 22 (10) EC Dual-Use Regulation:

The following purchased goods are subject to export controls. For the export of these goods from the Community an export license is necessary.”

For offers outside the European Union:

It is agreed, that the legally binding conclusion of this offer is subject to the suspensive condition that the competent authority grant the necessary export license.

ADDITIONAL PROVISIONS

In addition and subordinated to this quotation our General Conditions for Sales and Deliveries for International Business shall apply.

http://www.emag.com/Delivery_conditions_UN.pdf

<https://www.emag.com/de/iot-services.html>

<https://www.emag.com/media-recommendation/>

This document was created electronically and is valid without signature.

Sincerely,

EMAG Maschinenfabrik GmbH

Hartmut von Fugler
Managing Director

Do-Youn Koo
Area Sales Manager

Enclosures

EMAG-General conditions for sales and deliveries (UN)

Layout

Inner Race HBJ#27 (Processing plan, Timestudy, Quality commitments)

Inner Race HDJ#27 (Processing plan, Timestudy, Quality commitments)

EMAG-Interface to external handling systems

Price does not include options (*)

Offer : 200905 / 3

Hyundai WIA Corp

19/08/2022

33/33

Managing Directors:
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Michael Lanik, Sven Hartwich
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Tax No.: 63002/12866 FA Göppingen
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UniCredit Bank AG / Hypovereinsbank Stuttgart *IBAN DE02 6002 0290 0322 6388 17* BIC HYVEDEMM473
Commerzbank AG Tübingen *IBAN DE09 6414 0036 0890 9103 00* BIC COBADEFF641
Kreissparkasse Göppingen *IBAN DE96 6105 0000 0015 5885 93* BIC GOPSDE6GXXX
Bayerische Landesbank München *IBAN DE59 7005 0000 0004 2569 67* BIC BYLADEMMXXX